Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/20/2006
PI Name:	Buckey, Jay C. M.D.		
Project Title:	Otoacoustic Hearing Assessment of Space Station Crews		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Lebanon	**State:**	NH
Zip Code:	03756-0001	**Congressional District:**	2
Comments:	Address updated 9/2008		
Project Type:	Flight	**Solicitation / Funding Source:**	SMO 011
Start Date:	03/09/2005	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
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Flight Program:	Shuttle/ISS		
Flight Assignment:	In flight development phase (not yet manifested)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kline-Schoder, Robert (Creare, Inc.)		
Grant/Contract No.:	NNJ05HA65A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Hearing tests from the Salyut, Mir and Shuttle-Mir programs have shown temporary and, in some cases, permanent hearing threshold shifts after long-duration space flight. Currently, some areas on the International Space Station (ISS) do not meet noise standards. As a result, hearing conservation countermeasures have been implemented inflight for the astronauts. A missing component in the hearing conservation program is a reliable way to test hearing in-flight. Current methods for testing hearing while inflight have been shown in the past to be impractical due to high background noise levels. The purpose of this experiment is to develop and verify new hardware and protocols to assess and monitor hearing inflight using both distortion product otoacoustic emission (DPOAE) and audiometric hearing evaluations. Ground testing will be conducted to determine (1) the required DPOAE testing protocols for monitoring hearing in noisy environments, (2) whether background noise, like that present on ISS, renders pure-tone audiometry ineffective and (3)		

	the feasibility of making reflectance measurements to record changes in the middle ear which can affect DPOAE and audiometric measurements. Develop objective, sensitive, noise-tolerant hearing testing methods that can be used in space. Since astronauts work in a noisy environment, hearing testing is needed to insure that their hearing is not being damaged.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Noise-induced hearing loss is a significant problem in a variety of occupational settings and with aging. Reliable, noise-tolerant hearing testing methodologies could be used to assess hearing in these populations.
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 05/15/2026)